

Voter Proportion Outcomes

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Introduction

The total number of voters is a powerful reflection of how deep citizens are engaging in their community and country. High voter turnout sends a message about the health of a democracy, drawing the attention from political analysts and media outlets, which leads us to wonder what exactly makes people show up?

In some counties, voter turnout represents civic duty. There is relentless campaigning, ads, door knocking, and community discussion. In others, despite the fanfare, voter participation remains low. So what exactly drives people to cast their ballots? What traits do the people in these counties share? Could it be income, education, an urban or rural setting? Or maybe the presence of a dominant industry influencing the connection between people and politics?

In a study from the National Library of Medicine on the income gap in voting, numerous studies on Western democracies are cited and conclude that the less income an individual has, the less likely they are to vote (“The Income Gap in Voting,” NLM, 2022). We will test this claim in a detailed way, focusing on the attributes of county data for the 2024 general elections. The Bureau of Labor Statistics references findings from 2013 that found an inverse relationship between voter turnout and both local wages and employment. This literature emphasizes that working allows less time to become politically active (“Does labor market activity affect voter turnout?” BLS, 2015). However, this source cites this trend excludes presidential elections, and we want to explore whether these findings can be translated to the general election. Finally, education and political participation are regarded as coexisting mechanisms, and a 2023 study claims that education cost inflation as well as disruptive events such as COVID have impacted the value and perception of education (“How Education Shapes Voter Turnout,” NLM, 2023). Using our recent data, we would like to look for correlations based on these generalizations at a local scale.

In this project, we will examine voter turnout proportion across Virginia counties and independent cities, exploring how municipal factors such as median household income, population density, labor force, and more might predict the level of civic engagement, and aim to answer our research questions:

- Does a county with higher median income have a higher voter proportion?
- Does a county with a larger labor force have a lower voter proportion?
- Does a county with a majority of the population having a higher education (at least a Bachelor’s degree) have a higher voter proportion?

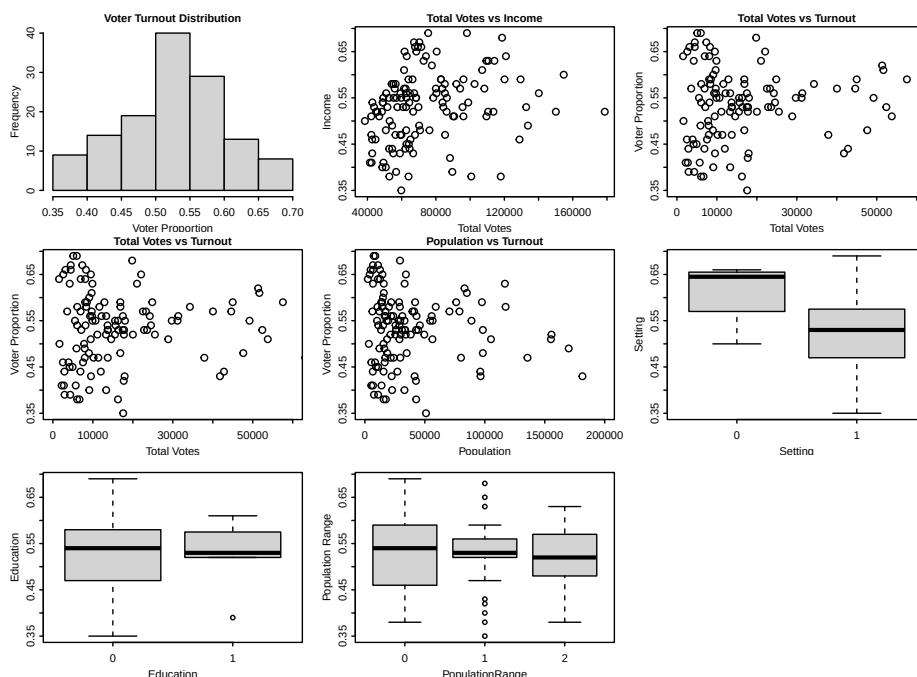
Data Summary

The dataset for this project was combined from the National Institute of Health (NIH), USDA, Bureau of Economic Analysis (BEA), Bureau of Labor Statistics (BLS), ODGA, Virginia Elections data as well as U.S. Census Bureau data. The median household income, population per county, and population density were collected from the National Institute of Minority and Health, including a median income for all ages, sex, and races between 2019-2023, and World Population Review. Following income, total voter turnout per county was taken as a sum of early, election day, absentee ballots, and other voting formats. This column's data is provided from the Virginia Department of Elections, a long-standing federal organization that holds recorded election results for descriptive purposes, which collected these counts for the 2024 general election. The voter proportion column was formed from a proportion of population and total voter turnout. Labor force, employment and unemployment data from the Virginia's office of Data Governance and analytics (ODGA) is also included to provide a better quantitative understanding of each counties' population makeup and provide more context.

Our qualitative variables' focus was to categorize as well as describe each county by its average education level, average weekly wage range, status as urban or rural, generalized population size and prominent industry type. The education variable was categorized by whether or not 50% of the county's citizens earned bachelors or higher, an average taken from the Department of Education for recording purposes for the year 2023. For average weekly wage, there are four ranges of consideration: \$949 and lower, \$950 to \$1099, \$1100 to \$1249, and \$1250 and higher. Urban or rural classification is based on the U.S. Census Bureau's description: To qualify as an urban area, the territory must encompass at least 2,000 housing units or have a population of at least 5,000. Population range categorized by small, medium and large is calculated using quartiles. Large included 54750 and higher, medium was from 27605.5, small was anything below. The last qualitative variable that was included, dominant industry, is sourced from the United States Department of Agriculture's department of Economic Research, where the counties typology code is formatted using descriptors such as high government, farming, manufacturing, and unspecified to describe each counties most influential employers. Because there were too many industry types, we manipulated this to two levels - if the county had a dominant industry or not. This assigned a number to each industry code for every county, which we then used to determine whether or not there was a prominent industry. This data sourced from 2025, is a part of a recording process that takes place for each county within the United States about every ten years. Its purpose is to describe regional economic and industry trends.

Our data was primarily sourced from government websites, so each observation cannot be publicly accessed or tampered with. Outliers that arise will most likely not be due to input error. To make qualitative variables we utilized a majority or non majority percentage for educated citizens per county, dominant industry, and population range. To do this we set a threshold and then used conditional formatting to write the qualitative data in. No counties were excluded for missing data.

To form the finalized data set, we added each data column with its coinciding county. To make sure there was not an input error on our end, we alphabetized the counties and matched each new column to its correct county. By using aggregated county-level data we protect individual privacy while still forming speculations to overall continue voting trends. Outliers such as Fairfax county which has a dense population were flagged for further analysis and verification.



EDA Summary

We began the explanatory data analysis by plotting a histogram of our response variable, which depicts the voter turnout proportion for each county in our data set, ranging from 35 to .75 percent. The majority of the data lies at about the .50 mark. Meaning a significant part of the population in each county can be suspected to have cast a vote. The distribution of the response variable is expressed as unimodal with very little to no skew.

Each scatterplot had the response variable Voter Proportion on the y-axis and the explanatory variable on the x axis. For most of the quantitative variable graphs, many of the points seemed to have condensed one side of the graph, with a few points significantly farther from the rest. However, if we zoom in, points are spread out and a slight positive linear relationship can be seen, with the strongest variable being Income. These graphs indicated there seemed to be some type of relationship between the variables and voter proportion.

The box plots represent the distribution of voter proportion across our qualitative variables. For the Setting variable the median voter proportion for rural living was noticeably higher than that of urban. Setting also exhibits a tighter interquartile range (IQR: $\sim .50-.65$) compared to Urban areas (IQR: $\sim .45-.60$), suggesting less variability in voter turnout. The Education variable, Population Range, and Dominant industries show a less drastic difference in median voter proportion. The education boxplot shows that counties with a majority of higher-educated residents tend to have a lower median voter proportion than those without. While the interquartile range for less educated residents is wider (IQR $\sim .50-.65$) than counties with a majority of educated residents (IQR: $\sim .45-.55$), indicating a more variable turnout in less educated counties.

The population range variable shows that counties with a lower population range have a higher median vote proportion, small (IQR: $\sim .43-.53$), medium (IQR: $\sim .53-.57$), and large counties (IQR: $\sim .43-.53$). The dominant Industry variable shows overlapping IQR's, but the counties with the presence of a dominant industry have a higher median and narrower interquartile range indicating fewer outliers.

In the correlations, none of the variables seem to have a strong positive correlation with voter proportion. Everything except Income and Education variables had a negative correlation. There is a slightly moderate negative correlation in Population Density. Income had a slightly positive correlation that was stronger than Education.

Additionally, multicollinearity was a concern, with several variables having high VIFs and an average VIF over 3. LaborForce and Employment are especially high, due to the fact that $\text{LaborForce} = \text{Employment} + \text{Unemployment}$. Population and TotalVote also have extremely high VIFs, which means they have strong collinearity with other variables, potentially with each other or with VoterProportion. From the correlation matrix heatmap, we see that LaborForce, Employment, and Population are highly correlated with each other. We chose to remove the variables TotalVote and Employment in the refined model.

Overall, Voter proportion is suitable for regression because the histogram shows the data is normally distributed, unimodal, and symmetric. In explanatory variables, Income, Population, Labor Force, Setting, Education, Population Range, Dominant Industry are able

to help predict the voter proportion based on the scatterplots and VIFs. Additionally the stepwise summary kept in Population Density, Income, Setting, Population.

Methods and Analysis

Our analysis began with the construction of an initial multiple linear regression model which contained all potential explanatory variables. Following this, we assessed the model with residual analysis and the assumptions needed which includes looking at the residual plots and histograms, influence plots. Our first model included the variables Income, total votes, population, population density, labor force, employment, setting, education, population range, and dominant industry. The model building process included a stepwise regression with a p-value threshold of .2 for entry and removal, followed by observation of individual variance inflation factors. This iterative process led to the removal of employment and total vote variables, due to their intense correlation with other predictors in the model.

In our residual analysis, we examined various diagnostic plots to assess our model assumptions. The leverage plots show a fanning-out pattern, suggesting heteroscedasticity, or non-constant variance of residuals. This pattern may be amplified by the presence of large outliers, which compress the remaining points and obscure the overall spread. Aside from the qualitative variables, the residuals generally center closely around zero, indicating that the linearity assumption is reasonably met. The Cook's Distance plot highlights three influential observations points 38, 69, and 88 that may be disproportionately affecting the model's fit and should be investigated further. For the studentized residuals it identifies the outliers and in our data it includes points 8, 38, 75, 76, and 88. This shows the standard deviations of the residuals and these show the more extreme residuals that are influential to the model.

The deleted studentized residuals graph also shows what outliers there are outside of the threshold of the absolute value of two which are 57, 69, 75, 76, 78. The QQ plot shows that the observations fall in a S shape to the line with outliers on the left, indicating a left skew that is reinforced by the histogram of residuals. To address the violations of regression assumptions we implemented a Box-Cox transformation. The procedure identified an optimal power transformation of $\lambda = 1.666667$. The application of the transformation to our response variable noticeably improved normality with a Shapiro-Wilk p-value = .3125.

The mean equal zero assumption was addressed early in the residuals vs fitted plot. The independence assumption was not a concern due to our data not being time series data and each observation not being dependent of another. To address Normality, the response

variable was transformed using a Box-Cox technique, and a Shapiro-Wilk test was used to verify its final normality assumption.

Results

Our Multilinear Regression analysis reveals that the most impactful predictors in our final model are Income, Unemployment, and the Setting variable, which identifies Urban vs. Rural counties. The final model is

$$\begin{aligned}\hat{y} = & 0.3733 + 1.168 \times 10^{-6} \cdot \text{Income} + 6.717 \times 10^{-7} \cdot \text{Population} + 1.922 \times 10^{-6} \cdot \text{LaborForce} \\ & - 1.292 \times 10^{-4} \cdot \text{Unemployment} - 0.09570 \cdot \text{Setting} - 0.05185 \cdot \text{Education} \\ & + 0.001207 \cdot \text{PopulationRange} + 0.009809 \cdot \text{DomIndustry}\end{aligned}$$

The final model illustrates that for every \$10,000 increase in median household income, the transformed variable shows a significant increase. Specifically, the variable with the most significance is income, meaning that for every \$1 increase in income, the transformed voter proportion increases by 0.000001466 units, holding all other variables constant. Although this is statistically significant, it is unrealistic to assume that higher income is directly linked to higher voter participation. Another notable variable was unemployment, as it had a negative impact, with every 1% increase in unemployment corresponding to a .000013 unit decrease in transformed turnout. To make sure the model is adequate the model was tested. As shown by a Global F test, the p-value of 0.0006884 is less than 0.05, making the model adequate. As for the individual t-test p values, only Income and Setting are significant. However, this does not completely disregard the rest of the variables. The Adjusted R squared value was marginally higher but it was similar to the multiple R-squared. After screening and testing the model multiple times, the current model is the best one and will be transformed with a different technique. The final model is to predict the voter proportion for a county in Virginia. Predictors include the income, population, labor force, setting, education, population range, and dominant industry. The estimated voter proportion for each county in Virginia is 3.733×10^{-1} .

There were five non-significant predictors according to their p-value for t-test. This includes labor force, education, dominant industry, population range, and population. For each one person increase in labor force, the predicted voter proportion total is expected to increase by 1.922×10^{-6} . For each county in Virginia that has a majority of people who have higher education, the voter proportion will decrease by 5.185×10^{-2} . For each county in Virginia that has a dominant industry, the voter proportion will increase by 9.809×10^{-3} . For each

population range, with 2 being large, 1 being medium, and 0 being small, there will be an increase of 1.207×10^{-3} . For each one person increase in Population, the predicted voter proportion total is expected to increase by 6.717×10^{-7} . These interpretations all have the assumption that the given explanatory variables were held constant. They were mainly kept in to keep our minimum of eight explanatory variables in the regression model equation to help transform the model for box-cox. However, some of the initial variables such as employment and average weekly wage were removed because they had a strong multicollinearity with some of the other variables that made the accuracy of the model to decrease.

The overall model explained 17.8% of the variance in turnout with an adjusted R^2 of .178; this model is also considered statistically significant with an F statistic of 4.5646. The adjusted R^2 value from this model is being heavily influenced by the binary variables within our dataset. Education and Industry dependent as binary variables oversimplify many socioeconomic dynamics within counties. To address non-normality in residuals we applied the Box-Cox transformation ($\lambda = 1.66666$). The positive influence of this transformation was confirmed by the p-value of the Shapiro-Wilk test's p-value $> .05$. Following our decision to reject the null and acknowledge our model as statistically significant, we conducted a Shapiro-Wilk test. This test yielded a p-value greater than the alpha threshold of .05, enabling us to reject the null hypothesis and accept the positive impact of the Box-Cox Transformation on the model's significance. By addressing these distributional challenges with this transformation we can adequately identify relationships with our explanatory and response. The low R^2 also points towards the opportunity to incorporate additional variables that further support the context of our research.

Conclusion

Ultimately, this regression model has some significance to it. There seemed to be a relationship with some variables based on their statistical significance in predicting voter proportion. This was especially true for income, unemployment, and setting. It suggested that counties with a larger median income, lower unemployment rates, and being in a rural area tend to have higher voter turnouts. This aligns with existing research showing that economic stability and community structure can influence civic engagement. While variables such as population, labor force, education, population range, and dominant industry did not have as much of a significant impact, their inclusion contributed to the overall model fit and allowed for the response variable to meet the normality assumption through a Box-Cox transforma-

tion. In the equation, voter proportion is equal to

$$\begin{aligned}\hat{y} = & 0.05374 + 1.294 \times 10^{-6} \cdot \text{Income} - 9.366 \times 10^{-7} \cdot \text{Population} + 1.442 \times 10^{-6} \cdot \text{LaborForce} \\ & - 8.880 \times 10^{-2} \cdot \text{Setting} - 4.039 \times 10^{-2} \cdot \text{Education} \\ & - 4.190 \times 10^{-3} \cdot \text{PopulationRange} + 2.619 \times 10^{-3} \cdot \text{DomIndustry}\end{aligned}$$

This regression model is not the most useful, but has some valid considerations included in the variables. To show how this model is not completely accurate, the example of Accomack County with \$57,500 of income, 33271 population, 15114 labor force, 546 unemployed, urban area, no majority of higher education, medium population range, and no dominant industry. The answer is approximately 32.7%, with the actual value being 52.5%. No majority of education and no dominant industry simplifies the several levels that contribute to this possible outcome which is why it may not be as accurate as it could be. A notable area of improvement is the further addition of contextual variables in future research. Our model has proven to be significant in a statistical sense, but practicality is not its strongest attribute. The variable that describes median household income has a trend indicating statistical significance. The capabilities of the model are limited due to the other variables that do not show as much statistical significance, especially the qualitative variables that are limited to two levels such as the dominant industry. This can also be improved by increasing the dataset to different counties all over the United States and exploring the data that has been collected over the years. This data was compiled from the last 5 years. Each column is no older than 2023, with exception to the population variable data that was compiled within the year range 2019-2023. The first limitation that occurs is this model being used to predict Voter proportion for the next presidential elections, this would be a severe extrapolation of our compiled data and model. The model would also not be able to predict voter proportions for any other state because of how small the sample data is from limiting it to Virginia counties. While only income and labor force research questions were predicted to impact voter proportion correctly, there is still evidence to suggest that there are many other factors that could impact voter turnout and increase or decrease voter proportion. Higher education may not have had a significant role in voter turnout in the model proposed, but it could still have other impacts in different sets of data. Future models could look into more data and variables that offer more insights into what affects voter proportions for different counties.

Appendix A: Data Dictionary

Variable Name	Abbreviated Name	Description
Income	Income	The median household income for Virginia by county over 2019 to 2023 in U.S. dollars.
TotalVote	Total Vote	Total number of ballots cast in November 2024 general election in Virginia by county.
Population	Population	The population of each Virginia county in 2025.
VoterProportion	Voter Proportion	The proportion of people who voted in each county taken by dividing the voter turnout (TotalVote) and population.
PopDensity	Population Density	Number of people per square mile in each county.
LaborForce	Labor Force	Annual average of the total civilian labor force (people who are working and actively looking for work) in each county in 2025.
Employment	Employment	Annual average of total number of people employed in each Virginia county in 2025.
Unemployment	Unemployment	Annual average of total number of people unemployed in each Virginia county in 2025.
Setting	Setting	Urban or rural status of each Virginia county.

Education	Education	The generalized education status of each Virginia county in 2023. (more than 50% have a bachelor's degree, less than 50% have a bachelor's degree)
PopulationRange	Population Range	The generalized population size of each Virginia county in 2025. (Small, medium $\geq$ 27605.5, large $\geq$ 54750)
DomIndustry	Dominant Industry	Status of a dominant industry existing in each Virginia county based on the 2025 ERS County Typology Codes.

Appendix B: Data Rows

	County	Income	TotalVote	Population		
1	Accomack County	57500	17475	33271		
2	Albemarle County	102617	67592	117338		
3	Alexandria City	113638	81594	155600		
4	Alleghany County and Clifton Forge City	55160	8344	14355		
5	Amelia County	66339	8070	13704		
6	Amherst County	67298	17510	31244		
	VoterProportion	PopDensity	LaborForce	Employment	Unemployment	Setting
1	0.53	74	15114	14568	546	1
2	0.58	163	61685	60089	1596	1
3	0.52	10373	101530	99043	2487	1
4	0.58	32	6041	5806	235	1
5	0.59	39	6550	6345	205	1
6	0.56	66	15331	14868	463	1
	Education	PopulationRange	DomIndustry			
1	0	1	0			
2	1	2	1			

3	1	2	1
4	0	0	1
5	0	0	0
6	0	1	1

Appendix C: Final Model Output and Plots

Call:

```
lm(formula = VoterProportion_transformed ~ Income + Population +  
    LaborForce + Unemployment + Setting + Education + PopulationRange +  
    DomIndustry, data = Final_Data_NoCounty)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.21535	-0.04750	0.01396	0.04364	0.16459

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.733e-01	4.458e-02	8.374	1.05e-13	***
Income	1.168e-06	3.645e-07	3.204	0.00173	**
Population	6.717e-07	1.221e-06	0.550	0.58324	
LaborForce	1.922e-06	1.801e-06	1.067	0.28798	
Unemployment	-1.292e-04	5.432e-05	-2.378	0.01896	*
Setting	-9.570e-02	3.902e-02	-2.453	0.01558	*
Education	-5.185e-02	3.050e-02	-1.700	0.09163	.
PopulationRange	1.207e-03	1.114e-02	0.108	0.91387	
DomIndustry	9.809e-03	1.430e-02	0.686	0.49419	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07544 on 123 degrees of freedom

Multiple R-squared: 0.2282, Adjusted R-squared: 0.178

F-statistic: 4.546 on 8 and 123 DF, p-value: 7.229e-05

Appendix D: References

Background

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Additional Help

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